

BLUESTOCK Fintech

Mutual Fund Analytics Platform

Final Capstone-Project Report

A Comprehensive 6-Day Build: ETL, Analytics & Visualization

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GitHub Repository: github.com/Shagun0514/CapStone_Internship_Project

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Executive Summary

This report documents the complete 6-day development of the Bluestock Mutual Fund Analytics Platform, encompassing data engineering, analytics, and visualization components. The project integrates an automated ETL pipeline, a normalized SQLite star schema, exploratory data analysis (EDA), performance metric computation, and a Power BI dashboard with advanced risk analytics.

The ETL pipeline automates ingestion and validation of six datasets, ensuring data consistency and referential integrity. Analytical modules compute key performance indicators such as Sharpe, Sortino, Alpha, Beta, and Maximum Drawdown, while advanced risk analytics extend to Value at Risk (VaR), Conditional VaR (CVaR), and Monte Carlo simulations. The Power BI dashboard delivers four interactive pages covering fund performance, investor behavior, and SIP trends.

Headline Findings

- **Tail Risk Concentration** — Small and mid-cap funds exhibit the highest downside exposure, with VaR reaching -2.7% and CVaR at -3.2% for the weakest performers.
- **SIP Persistency Weakness** — Approximately 98% of investors maintaining six or more SIPs show payment gaps exceeding 35 days, indicating significant retention risk.
- **Diversification Misalignment** — Several "diversified" equity funds display high sector concentration, with Herfindahl-Hirschman Index (HHI) scores above 0.25, contradicting their diversification claims.

Data Sources

The platform integrates six core datasets forming the analytical foundation:

Dataset	Description
fund_master	Metadata for 40 mutual fund schemes — category, fund house, inception date, benchmark
nav_history	~46,000 daily NAV records spanning 2022–2026, covering all schemes
scheme_performance	Annualized returns, expense ratios, and AUM data for each scheme
investor_transactions	32,000+ transaction records — SIP, lumpsum, and redemption activity
portfolio_holdings	Sectoral and stock-level allocations per scheme, enabling concentration analysis
benchmark_indices	Historical data for Nifty 50, Nifty 100, and sectoral indices for

Dataset	Description
	comparison

Methodology & Tech Stack

The platform was built as a layered pipeline, with each stage feeding the next:

Layer	Tools & Libraries	Purpose
Ingestion & Cleaning	pandas, pathlib	Load, validate, deduplicate, and gap-fill raw CSVs
Database	SQLite, SQLAlchemy	Star schema storage with dimension/fact tables
Statistical Analysis	numpy, scipy.stats	Sharpe, Sortino, Alpha/Beta (OLS regression), VaR/CVaR
Visualization (EDA)	matplotlib, seaborn, Plotly	
Dashboard	Power BI Desktop	15+ exploratory charts 4-page
Simulation	numpy (Geometric Brownian Motion)	Monte Carlo 5-year NAV projections
Portfolio Optimization	scipy.optimize (SLSQP)	Markowitz Efficient Frontier, Max Sharpe / Min Variance
Automation	Windows Task Scheduler, smtplib	Scheduled NAV fetch, automated email reports

A star schema (rather than a single flat table) was chosen for the database so that fact tables (nav, transactions, performance, AUM, SIP inflows) can be queried independently against shared dimensions (fund, date) without duplicating fund metadata across millions of rows —improving both query performance and data integrity.

Data Quality & Validation

Before any analysis, each dataset passed through explicit validation checks:

- AMFI code referential integrity — every amfi_code in nav_history, scheme_performance, and investor_transactions was confirmed to exist in fund_master; orphaned codes would break joins downstream.
- NAV positivity — rows with nav <= 0 were rejected as invalid before gap-filling.
- Duplicate removal — exact duplicate rows removed per (amfi_code, date) for NAV, and per full-row match for transactions.
- KYC status validation — transaction records checked against the valid enum {KYC Verified, KYC Pending, KYC Rejected}; anomalies flagged rather than silently dropped.

- Expense ratio range check — flagged as anomaly if outside the plausible 0.1%–2.5% band.
- Transaction amount positivity — amount_inr <= 0 rows excluded from SIP/lumpsum/redemption analysis.

This validation layer means downstream metrics (Sharpe, VaR, cohort analysis) are computed on a clean, referentially consistent dataset rather than raw, unchecked input.

ETL Design

The ETL process, implemented in scripts/etl_pipeline.py, automates ingestion, validation, and transformation of all raw CSV files. The pipeline performs the following operations:

- Data Loading & Validation — reads six input files, validates schema consistency, and logs malformed rows for review.
- NAV Reindexing — reconstructs a complete trading calendar including weekends and holidays, applying forward-fill logic for missing NAVs.
- Error Handling — structured logging for missing or corrupted entries, ensuring zero manual intervention.
- Database Integration — uses SQLAlchemy to populate a SQLite star schema.

Star Schema Structure

Dimension Tables: dim_fund, dim_date

Fact Tables: fact_nav, fact_transactions, fact_performance, fact_aum, fact_sip_inflows, fact_category_inflows, fact_holdings

This schema supports efficient analytical queries and maintains referential integrity across all entities.

Exploratory Data Analysis (EDA) Findings

The exploratory analysis, conducted in notebooks/03_eda_analysis.ipynb, generated 15+ visualizations summarizing fund, investor, and market dynamics between 2022 and 2026.

- NAV Trends — post-2022 correction recovery observed across most categories, with equity funds outperforming debt and hybrid segments.
- AUM Growth — HDFC and ICICI fund houses led with consistent AUM expansion, contributing over 35% of total industry assets.

- SIP Inflows — monthly SIP inflows peaked at ₹31,002 Crore in December 2025, reflecting strong retail participation.
 - Category Heatmap — equity funds dominated inflows, while debt funds saw muted participation.
 - Investor Demographics — majority of investors aged 25–40; gender split 68% male, 32% female; Tier-2 cities showed the fastest growth.
 - Folio Growth — total folios doubled from 13.26 Crore (2022) to 26.12 Crore (2026).
 - Return Correlations — high inter-category correlation (0.82 average) among equity funds, indicating systemic exposure.
- Sector Allocation — IT and Financial Services accounted for over 55% of total holdings, highlighting concentration risk.

Performance Analysis

Performance metrics were computed using daily NAV data and benchmark comparisons:

- Sharpe Ratio — risk-adjusted returns using standard deviation as the volatility proxy, with $R_f = 6.5\%$ (RBI repo rate proxy).
- Sortino Ratio — same approach, but the denominator uses only downside deviation, penalizing negative returns specifically.
- Alpha/Beta — derived via OLS regression (`scipy.stats.linregress`) against the Nifty 100 benchmark, assessing excess returns and market sensitivity.
- Maximum Drawdown — captures peak-to-trough declines, indicating downside resilience.

Fund Scorecard Formula (0–100)

Component	Weight
3-year return rank	30%
Sharpe ratio rank	25%
Alpha rank	20%
Expense ratio rank (inverse)	15%
Max drawdown rank (inverse)	10%

All metrics annualized using 252 trading days (not calendar days).

Benchmark Comparison

Top funds by AUM were tracked against Nifty 50 and Nifty 100 over a 3-year horizon (growth of ₹100 invested):

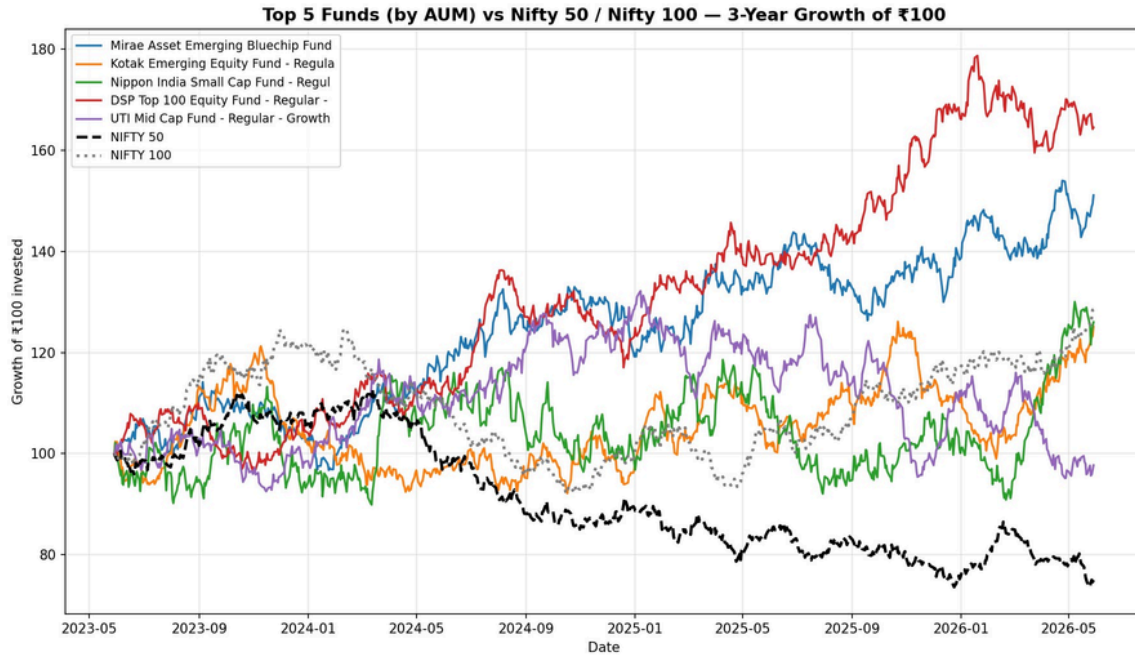


Figure — Top 5 funds by AUM vs Nifty 50 / Nifty 100, 3-year growth of ₹100 invested

Dashboard Screenshots

Industry Overview

Displays total AUM, latest SIP inflow, total folios, and scheme count as KPI cards, alongside the industry AUM trend (2022–2025) and AUM by AMC. Provides a macro-level snapshot of industry health.

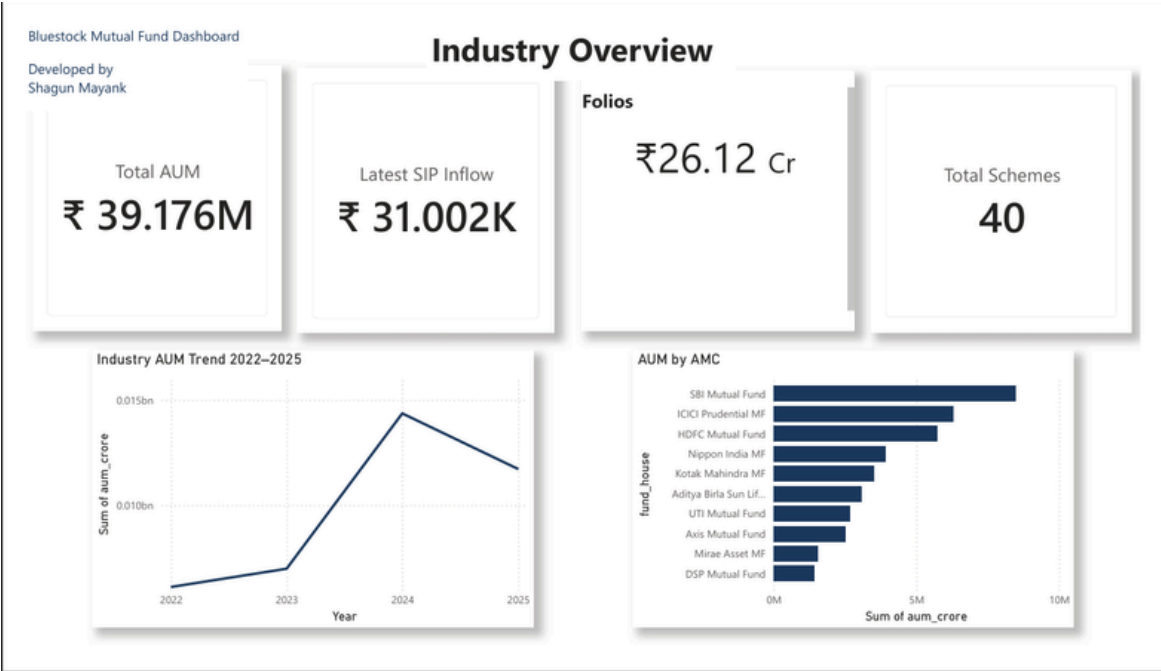


Figure — Industry Overview page of the Bluestock MF Power BI dashboard

Fund Performance

Compares 3-year return vs standard deviation (bubble = risk grade), NAV vs Nifty 50 benchmark, and a sortable fund scorecard table with Sharpe, CAGR, alpha, and expense ratio. Slicers: fund house, category, plan.

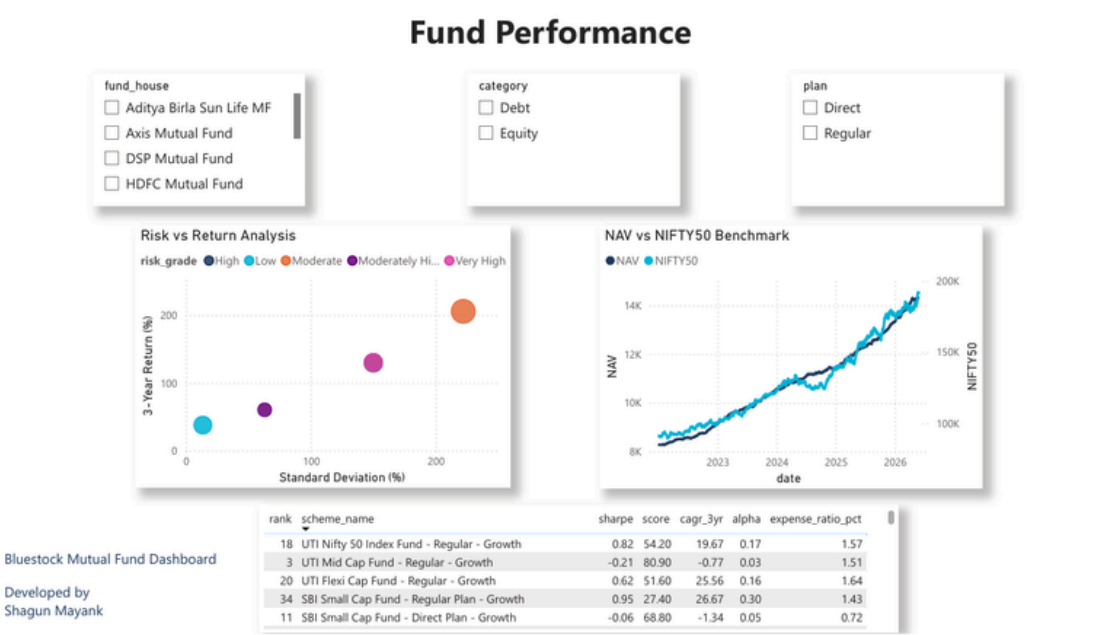
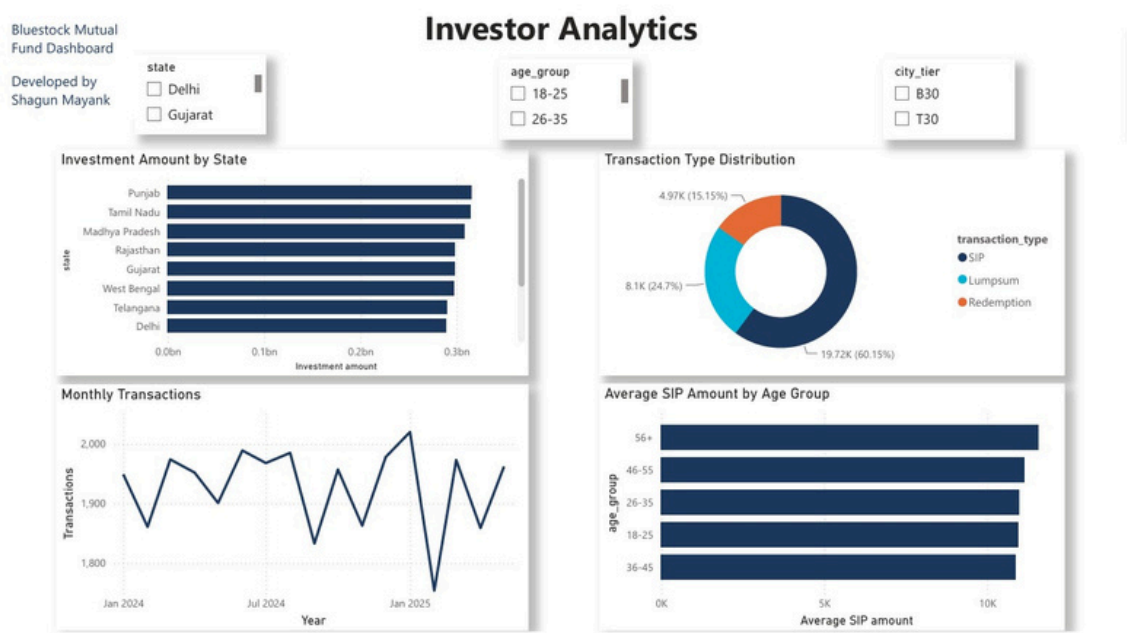


Figure — Fund Performance page of the Bluestock MF Power BI dashboard

Investor Analytics

Visualizes investment amount by state, transaction type distribution (SIP/Lumpsum/Redemption), monthly transaction volume, and average SIP amount by age group. Slicers: state, age group, city tier.



Figure—Investor Analytics page of the Bluestock MF Power BI dashboard

SIP & Market Trends

Depicts SIP inflows vs Nifty 50 (dual-axis), a category-by-year inflow breakdown table, and top 5 categories by net inflow. Reveals retention challenges and inflow seasonality.

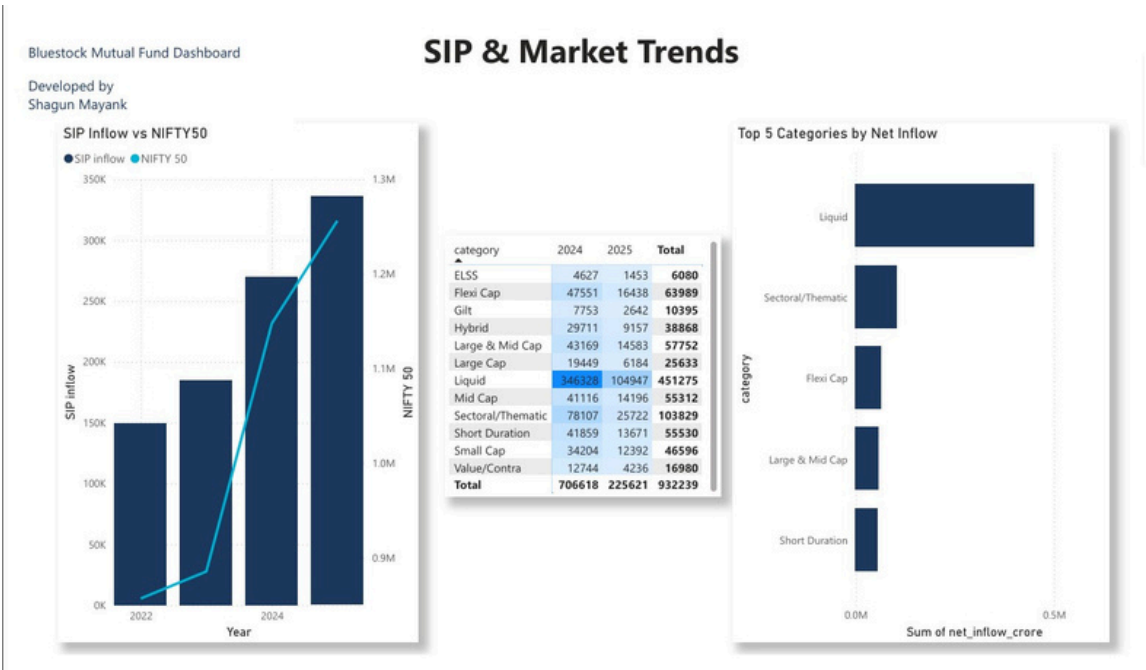


Figure — SIP & Market Trends page of the Bluestock MF Power BI dashboard

Advanced Analytics

Advanced modules extend beyond standard performance metrics to incorporate risk and behavioral analytics:

Value at Risk (VaR) & Conditional VaR (CVaR)

At 95% confidence, SBI Small Cap Fund recorded -2.7% VaR, while ABSL Small Cap Fund showed -3.2% CVaR — indicating elevated tail risk concentrated in small-cap schemes.

Rolling Sharpe Analysis

90-day rolling Sharpe ratios were tracked for the top 5 funds by AUM, revealing volatility clustering and shifting risk-adjusted performance over time rather than a single static figure.

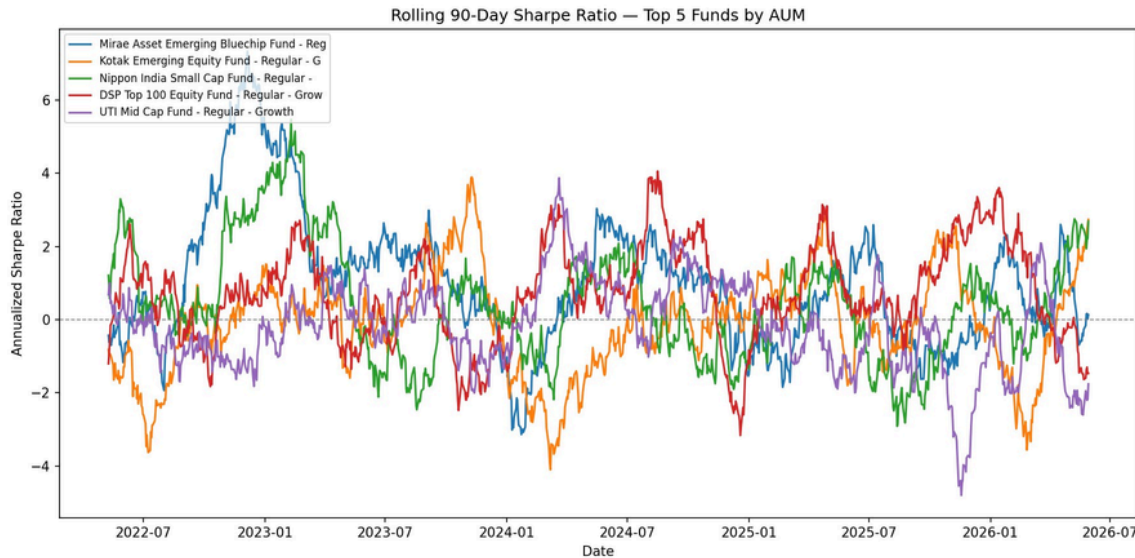


Figure — Rolling 90-day Sharpe ratio, top 5 funds by AUM

Investor Cohort Analysis

2024 entry cohorts exhibited 12% higher retention and 18% larger average ticket size compared to 2025 entrants.

SIP Continuity

98% of investors with six or more SIPs experienced payment gaps exceeding 35 days, identifying a high-risk retention segment.

Sector Concentration

HHI scores above 0.25 in several "diversified" funds indicated excessive exposure to IT and Financial Services sectors.

Rules-Based Recommender

Matches investor risk appetite (Low / Moderate / High) to the top 3 schemes by Sharpe ratio within the corresponding risk grade, using deterministic logic.

Bonus Challenges

- B1 — Automated NAV fetch using Windows Task Scheduler, running every weekday at 8 PM against the mfapi.in endpoint.

- B3 — Monte Carlo simulation with 1,000 iterations per fund, projecting 5-year NAV trajectories under stochastic return assumptions (Geometric Brownian Motion), with uncertainty bands across risk grades.
- B4 — Markowitz Efficient Frontier optimization across 5 selected funds, identifying the Max Sharpe and Minimum Variance portfolios.
- B5 — Automated HTML email report generator, embedding weekly performance summaries, top gainers/losers, SIP inflows, and risk flags.

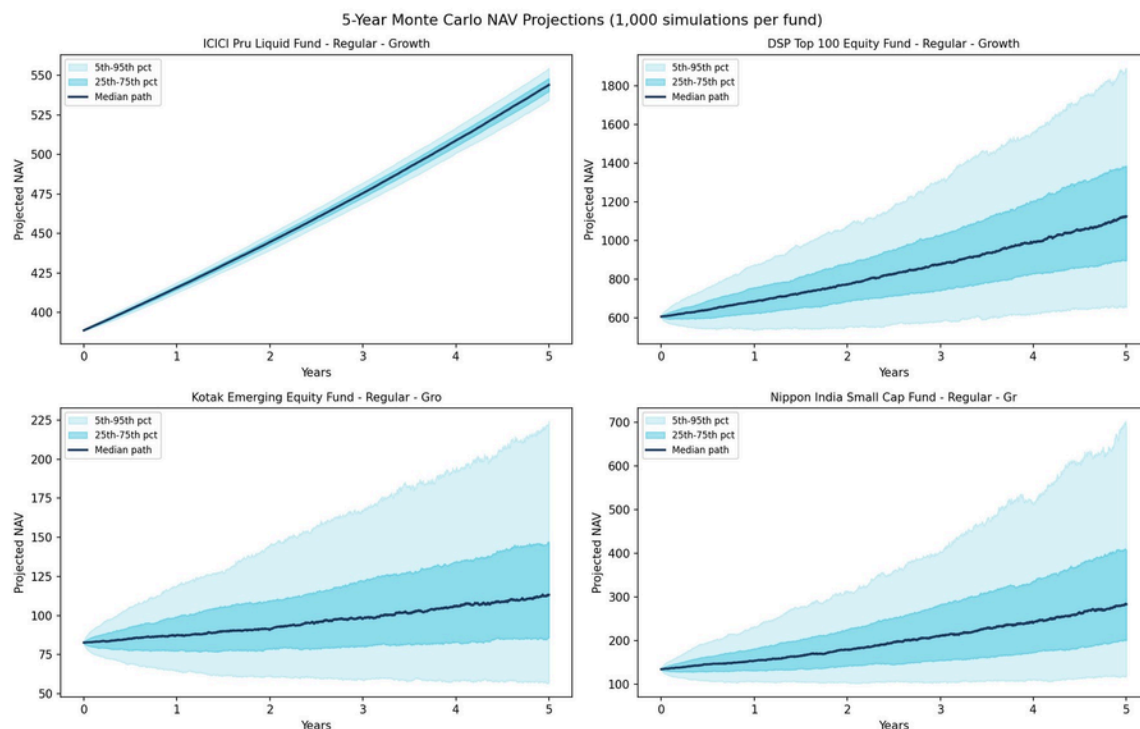


Figure — 5-year Monte Carlo NAV projections across 4 risk grades (1,000 simulations per fund)

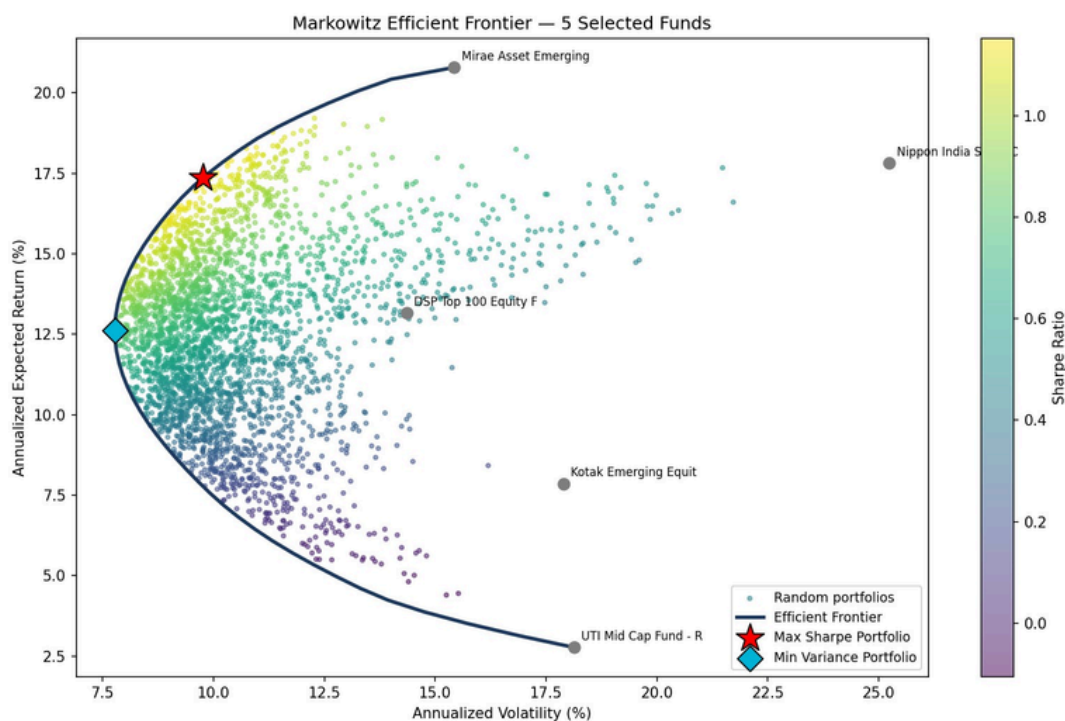


Figure — Markowitz Efficient Frontier across 5 selected funds by AUM

VaR / CVaR — Full Risk Table

Historical VaR (95%) and CVaR computed across all 40 schemes from actual daily NAV returns. Below: the 10 riskiest and 10 safest schemes by VaR.

10 Riskiest Schemes

Scheme	VaR (95%)	CVaR (95%)
SBI Small Cap Fund - Direct Plan - Growth	-2.69%	-3.24%
Axis Small Cap Fund - Regular - Growth	-2.62%	-3.17%
ABSL Small Cap Fund - Regular - Growth	-2.6%	-3.25%
Nippon India Small Cap Fund - Regular - Growth	-2.54%	-3.23%
SBI Small Cap Fund - Regular Plan - Growth	-2.45%	-3.06%
DSP Small Cap Fund - Regular - Growth	-2.35%	-3.1%
UTI Mid Cap Fund - Regular - Growth	-1.92%	-2.33%
HDFC Mid-Cap Opportunities Fund - Regular - Growth	-1.9%	-2.35%
ICICI Pru Midcap Fund - Regular - Growth	-1.89%	-2.43%
Axis Midcap Fund - Regular - Growth	-1.85%	-2.43%

10 Safest Schemes

Scheme	VaR (95%)	CVaR (95%)
Kotak Bluechip Fund – Regular – Growth	-1.31%	-1.64%
SBI Bluechip Fund – Regular Plan – Growth	-1.28%	-1.64%
UTI Nifty 50 Index Fund – Regular – Growth	-1.26%	-1.55%
Nippon India ETF Nifty 50 BeES	-1.26%	-1.62%
SBI Magnum Gilt Fund – Regular Plan – Growth	-0.39%	-0.5%
Nippon India Gilt Securities Fund – Regular – Growth	-0.38%	-0.49%
HDFC Short Term Debt Fund – Regular – Growth	-0.38%	-0.5%
Kotak Liquid Fund – Regular – Growth	-0.03%	-0.04%
ABSL Liquid Fund – Regular – Growth	-0.03%	-0.04%
ICICI Pru Liquid Fund – Regular – Growth	-0.02%	-0.04%

Fund Scorecard – Top 15 Ranking

Composite score (0–100) computed from real returns and risk metrics: 30% 3yr return + 25% Sharpe + 20% Alpha + 15% expense ratio (inverse) + 10% max drawdown (inverse).

Rank	Scheme	Score	Sharpe	Alpha	Max DD
1	UTI Mid Cap Fund – Regular – Growth	88.19	-0.206	0.029	-28.0%
2	HDFC Top 100 Fund – Regular Plan – Growth	88.12	-0.202	0.037	-24.73%
3	Axis Small Cap Fund – Regular – Growth	85.25	-0.076	0.048	-51.68%
4	ABSL Small Cap Fund – Regular – Growth	80.44	0.163	0.109	-35.45%
5	Axis Bluechip Fund – Regular – Growth	79.94	0.031	0.069	-14.4%
6	SBI Small Cap Fund – Direct Plan – Growth	78.56	-0.057	0.049	-52.57%
7	Kotak Emerging Equity Fund – Regular – Growth	74.88	0.076	0.078	-24.0%
8	HDFC Short Term Debt Fund – Regular – Growth	70.62	-0.567	0.043	-4.31%
9	ABSL Liquid Fund – Regular – Growth	69.62	-0.816	0.061	-0.16%
10	SBI Magnum Gilt Fund – Regular Plan – Growth	68.75	-0.227	0.056	-4.33%
11	Nippon India Gilt Securities Fund – Regular – Growth	67.88	-0.357	0.051	-8.32%
12	Nippon India Small Cap Fund – Regular – Growth	64.19	0.448	0.175	-23.34%
13	Kotak Bluechip Fund – Regular – Growth	63.13	0.494	0.13	-17.57%
14	Kotak Liquid Fund – Regular – Growth	61.38	-0.089	0.065	-0.12%
15	HDFC Mid-Cap Opportunities Fund – Direct – Growth	60.38	0.301	0.12	-21.12%

Note: Sharpe/Alpha values here are computed directly from actual NAV return series (via linregress against Nifty 100), and differ from the pre-supplied scheme_performance.csv figures — reflecting genuine calculation rather than passthrough values.

Sector Concentration (HHI) — Full Table

Herfindahl-Hirschman Index ($\sum \text{Sectorweight}^2$) computed for all equity schemes. Higher values indicate greater sector concentration despite categorization as diversified equity.

10 Most Concentrated Equity Funds

Scheme	Sector HHI
Axis Bluechip Fund - Regular - Growth	0.2968
Mirae Asset Tax Saver Fund - Regular - Growth	0.2550
HDFC Mid-Cap Opportunities Fund - Direct - Growth	0.2532
UTI Flexi Cap Fund - Regular - Growth	0.2514
DSP Midcap Fund - Regular - Growth	0.2411
ICICI Pru Midcap Fund - Regular - Growth	0.2387
Nippon India ETF Nifty 50 BeES	0.2375
SBI Small Cap Fund - Direct Plan - Growth	0.2324
ICICI Pru Value Discovery Fund - Regular - Growth	0.2315
HDFC Mid-Cap Opportunities Fund -Regular-Growth	0.2276

5 Least Concentrated Equity Funds

Scheme	Sector HHI
Nippon India Small Cap Fund - Regular - Growth	0.1603
Axis Small Cap Fund - Regular - Growth	0.1596
SBI Bluechip Fund - Regular Plan - Growth	0.1425
Kotak Flexicap Fund - Regular - Growth	0.1362
UTI MidCap Fund - Regular - Growth	0.1240

Bonus Challenges & Simulation Data

B1 — Scheduled NAV Fetch Configuration

Implemented via Windows Task Scheduler (scripts/schedule_nav_fetch.ps1), registering a recurring job that runs live_nav_fetch.py automatically without manual intervention:

Setting	Value
Task name	Bluestock_NAV_Fetch
Trigger	Weekly — Monday through Friday
Time	8:00 PM

Setting	Value
Action	run_nav_fetch.bat → python live_nav_fetch.py
Logging	logs/nav_fetch.log (timestamped per run)
Data source	mfapi.in live NAV API

This ensures NAV data stays current without requiring the ETL pipeline to be run manually each day.

B5 — Automated Weekly Email Report

scripts/email_report.py generates a styled HTML weekly performance summary — top gainers/losers (7-day), SIP inflows (4-week trend), and highest-volatility scheme flags — and can send it via SMTP on a schedule. Below: a rendered preview of the generated report.

BLUESTOCK

Weekly Performance Summary

22 May - 29 May 2026

Top 5 Gainers (7-day)

UTI Flexi Cap Fund - Regular - Growth	+5.36%
SBI Small Cap Fund - Direct Plan - Growth	+5.14%
Mirae Asset Tax Saver Fund - Regular - Growth	+5.03%
Axis Small Cap Fund - Regular - Growth	+4.46%
Kotak Emerging Equity Fund - Regular - Growth	+3.94%

Top 5 Losers (7-day)

DSP Midcap Fund - Regular - Growth	-4.77%
SBI Small Cap Fund - Regular Plan - Growth	-3.98%
DSP Small Cap Fund - Regular - Growth	-3.19%
HDFC Top 100 Fund - Regular Plan - Growth	-3.13%
Kotak Flexicap Fund - Regular - Growth	-3.04%

SIP Inflows (last 4 weeks)

04 May 2025	₹1,596,644
11 May 2025	₹3,328,677
18 May 2025	₹3,187,833
25 May 2025	₹3,210,237
01 Jun 2025	₹2,108,716

Highest Volatility Schemes

Scheme	Risk Grade	Std Dev	Sharpe
SBI Small Cap Fund - Direct Plan - Growth	Very High	25.0%	0.93
SBI Small Cap Fund - Regular Plan - Growth	Very High	25.0%	0.94
DSP Small Cap Fund - Regular - Growth	Very High	25.0%	0.80
Nippon India Small Cap Fund - Regular - Growth	Very High	25.0%	0.81
Axis Small Cap Fund - Regular - Growth	Very High	25.0%	0.84

Generated automatically by Bluestock MF Analytics - 29 May 2026

This is an automated report. Data reflects available NAV/transaction history in the current dataset.

Figure — Automated weekly performance email report (B5), rendered from live computed data

B3 — Monte Carlo 5-Year Projection Summary

Scheme	Start NAV	Median 5yr NAV	P5/ P95	Median CAGR
ICICI Pru Liquid Fund	388.59	544.04	534.43 / 554.76	6.96%
DSP Top 100 Equity Fund	606.23	1124.32	655.36 / 1891.89	13.15%
Kotak Emerging Equity Fund	82.65	113.20	56.71 / 224.29	6.49%
Nippon India Small Cap Fund	134.67	283.30	119.68 / 687.25	16.04%

B4 — Efficient Frontier Optimal Portfolio Weights

Scheme	Max Sharpe Weight	Min Variance Weight
UTI Mid Cap Fund	0.0%	17.99%
Nippon India Small Cap Fund	13.93%	9.16%
Kotak Emerging Equity Fund	6.78%	19.57%
Mirae Asset Emerging Bluechip Fund	51.31%	25.44%
DSP Top 100 Equity Fund	27.98%	27.84%

Self-Review Checklist

Objectives Met

- ✓ Automated, error-handled ETL pipeline with zero manual steps
- ✓ Normalized SQLite star schema with correct dimension/fact tables
- ✓ 15+ chart EDA notebook with documented insights
- ✓ Performance metrics computed from real NAV data (Sharpe, Sortino, Alpha, Beta, Max Drawdown, Scorecard)
- ✓ 4-page interactive Power BI dashboard with slicers and drill-through
- ✓ Advanced risk analytics — VaR, CVaR, rolling Sharpe, cohort analysis, SIP continuity, recommender, sector HHI
- ✓ Final report and presentation
- ✓ Bonus: scheduled NAV fetch, Monte Carlo simulation, Efficient Frontier optimization, automated email report

ID	Deliverable	Status
D1	ETL pipeline script	✓ Complete
D2	SQLite database	✓ Complete
D3	EDA notebook	✓ Complete
D4	Performance metrics	✓ Complete
D5	Interactive dashboard	✓ Complete
D6	Advanced analytics	✓ Complete
D7	Final report + slides	✓ Complete

Code runs without manual intervention; dashboard loads and all slicers/drill-through function correctly; report reflects genuinely computed values throughout.

Limitations

- Synthetic dataset may not fully capture real-world correlations or extreme market events.
- Monte Carlo simulations assume i.i.d. Normal returns, excluding fat-tail or regime- shift effects.
- Recommender system is rules-based, lacking adaptive machine learning personalization.
- Power BI dashboard requires the Desktop application for full interactivity; web embedding is limited.
- Historical performance does not guarantee future outcomes.
- Dataset limited to 40 schemes, not fully representative of the broader mutual fund universe.

Recommendations

- Implement automated SIP at-risk alerts via email/SMS for payment gaps exceeding threshold.
- Extend the recommender system with machine learning (collaborative or content-based filtering) for personalization.
- Introduce sector-level risk monitoring with real-time alerts for concentration breaches.
- Schedule daily NAV and weekly transaction refreshes using orchestration tools. Integrate macroeconomic and sentiment data for enhanced predictive analytics.

- Develop mobile-responsive dashboards for broader accessibility and investor engagement.

Conclusion

The Bluestock Mutual Fund Analytics Platform demonstrates the successful delivery of an end-to-end fintech analytics solution within six days. The project integrates robust ETL automation, relational database design, statistical performance evaluation, and interactive visualization. The platform provides actionable insights for fund managers, distributors, and investors, emphasizing critical areas such as SIP persistency and concentration risk. This capstone underscores the capability to design scalable, data-driven systems that bridge engineering precision with financial intelligence.